

Abstract

This project develops a basketball modeling framework to answer a fundamental question: what is winning basketball? The objective is to identify the statistical advantages that consistently separate winning teams from losing teams, and to translate those advantages into a predictive model of future team performance. The framework begins with historical game data, using XGBoost classifiers and SHAP analysis to identify which box score categories produce the largest, most consistent shifts in predicted win probability. I found that differential statistics produced the strongest SHAP results, and partial dependence plots were then used to narrow them to a final feature set. The final model classified game outcomes with 88.7% accuracy and a Brier score of 0.0823, while pace showed minimal predictive value. To project future performance, the model was applied to current rosters to estimate per-game statistics across the upcoming NBA schedule, supplemented by a rookie-projection model based on 2K ratings and adjustments for back-to-backs and home-court advantage.

The teams the model favored most heavily tended to be those with the most star talent and those expected to shoot many three-pointers and defend well. The resulting win projections differed from Vegas preseason win totals by an average of 7.4 wins per team and fell within 3.5 wins of the market line for 11 of 30 teams. These results suggest that differential, relative team performance is a strong basis for modeling team success, though the projection pipeline's assumptions leave room for refinement.

1. Introduction

Predicting the outcome of a basketball game, or a team's win total over the course of a full season, has attracted attention from both the sports industry and researchers for decades. From sportsbooks setting betting lines to NBA front offices evaluating rosters and strategies, organizations have a strong incentive to understand which factors most consistently contribute to winning.

This project began with a narrower question: Which box score statistics contribute to winning NBA games? Basketball, like most competitive sports, is ultimately played to win, and teams commonly assume that certain aspects of performance, such as efficient shooting, strong defense, and effective rebounding, translate into more victories. This project tests those assumptions directly by examining a broad range of box score statistics and determining whether winning can be explained by a relatively small set of measurable statistical advantages.

Understanding these relationships has practical implications for team building, coaching, and performance evaluation. If a small set of statistical advantages can reliably distinguish winning teams from losing teams, teams can use these findings to inform roster construction and strategic decision-making. This approach is similar to commonly held beliefs in sports, such as the baseball idea that “small ball wins games” or the basketball belief that “defense wins championships.” Rather than relying solely on conventional wisdom, this project seeks to test these assumptions using data.

The project was divided into two primary objectives:

1. Identify the statistical drivers of winning. Alongside cleaning the data, the box score statistics were organized into broader categories to determine which features within those categories were most strongly associated with winning.
2. Project and simulate future team performance. After identifying the most informative features and developing a predictive model, I projected the box score statistics of two teams for an upcoming season based on their updated rosters. I then compared their projected performance to generate simulated season standings.

Together, these objectives examine both sides of the problem: which statistical characteristics are associated with winning and whether those relationships can be used to project team performance in a future season.

2. Background

Predicting the outcome of an NBA game is difficult, even for established betting markets. According to Zukiewicz (2023), pregame point spreads correctly identified the winner in approximately 65% of NBA games during the 2022–23 season and explained roughly 17% of the variance in final scoring margin. These results illustrate both the difficulty of predicting basketball outcomes and the extent to which commonly used forecasting methods can capture meaningful information about team strength. Despite this difficulty, researchers and analysts have spent years developing models that attempt to estimate the probability of winning as accurately as possible.

One example is Michael Beuoy’s model from *Inpredictable*, which uses locally weighted logistic regression to estimate in-game win probability. The model incorporates score margin, time remaining, possession, and the pregame point spread as a proxy for team strength. Zukiewicz (2023) developed an alternative model using a small neural network and incorporated additional variables, including free throws, ejections, and bonus status. He reported an average Brier score of 0.152 compared with 0.163 for the unpredictable model. However, Zukiewicz

notes that the models were evaluated on different sets of games, meaning that the comparison is not a direct head-to-head evaluation.

Other analysts have approached basketball forecasting by incorporating measures of team strength and situational factors. Hoops Junkie, for example, has explored factors such as Elo ratings, seasonal regression, and the effects of back-to-back games and rest. These approaches generally combine a pregame estimate of team strength with information that becomes available during the game, such as the current score and time remaining. However, these models may not fully account for factors such as which players are on the court, individual player performance, foul trouble, or coaching decisions.

Much of the existing basketball forecasting literature therefore focuses on estimating win probability at the team or game level, particularly before or during a game. While these models are useful for predicting individual game outcomes, they do not directly address how changes to a team's roster may translate into expected performance over an entire upcoming season.

2.2 Player Projection Systems in Baseball

Outside of basketball, baseball has a particularly mature tradition of translating individual player performance into team level win projections. Sabermetric projection systems such as PECOTA, ZiPS, and Steamer have spent years refining methods for forecasting individual player production and aggregating those forecasts into team level expectations.

PECOTA (Player Empirical Comparison and Optimization Test Algorithm), originally developed by Nate Silver from Baseball Prospectus, projects player performance by comparing a player's statistical profile and career trajectory with those of historical comparables. Rather than producing a single deterministic prediction, the system generates a distribution of potential outcomes. ZiPS incorporates factors such as multi-year weighted performance, aging curves, and historical player comparables, while Steamer relies more heavily on regression-based methods and regression toward the mean.

These systems provide a useful framework for thinking about roster-based forecasting. Rather than simply evaluating how strong a team was in the previous season, player projection systems attempt to estimate how individual players will perform in the future and then use those projections to construct an expectation for the team as a whole. This creates a relatively transparent pipeline from individual player performance to a projected team performance and expected wins.

2.3 Research Gap

This project attempts to bridge the two approaches together. Rather than predicting individual games exclusively through pregame or in-game information, the model first identifies

the box score statistics most strongly associated with winning and then uses that information to establish a statistical profile for each team. Subsequently, the model projects how two teams may perform based on their updated rosters and uses those projections to simulate their performance over a season.

In this way, the project combines elements of both basketball forecasting and baseball projection systems. It uses winning related team statistics as the foundation of the model while incorporating the broader idea of projecting future performance from roster composition. The goal is not simply to predict individual game outcomes, but to examine whether relationships between box score statistics and winning can be used to construct a data-driven framework for projecting team performance over an upcoming NBA season.

3. Methodology

The modeling framework consists of two stages. Stage 1 develops a game-level model that estimates win probability from team box-score statistics, while Stage 2 uses player-level projections to construct future team profiles and simulate an upcoming NBA season. Detailed data-processing procedures, feature engineering, model specifications, and full results are provided in the supplemental notebooks.

3.1 Stage 1: Game-Level Win Model

I used team box score data for NBA games from Moore (n.d.), combining standard and extended team statistics files. Games with missing extended statistics, malformed records, or duplicate records were removed. Game labels, including All-Star Games (ASG), play-in games, regular season games, and playoff games, were categorized by game type.

Each game was restructured into two team perspective observations, with each observation representing one team's performance relative to its opponent. Team - opponent differentials were calculated for each statistic. Only regular season games from 1996 onward were used to train and evaluate the primary model, while playoff games were held out as a separate evaluation set to assess generalization to a different game environment. Regular season games were randomly divided within each season into training, validation, and test sets using an approximately 70/20/10 split.

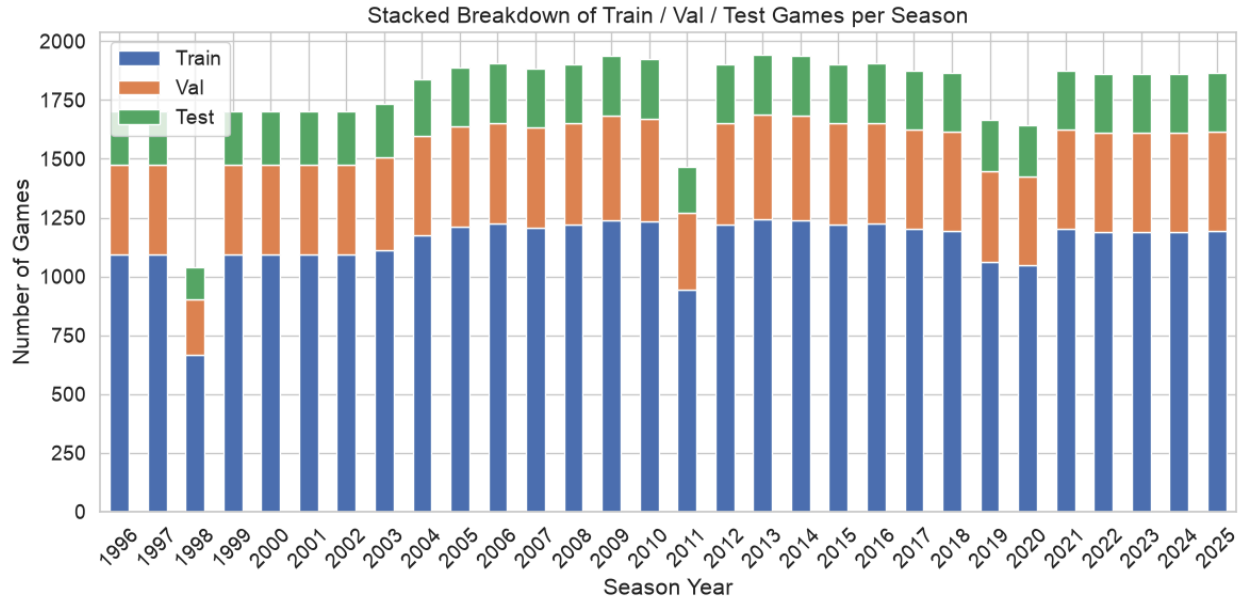


Figure 1: Stacked breakdown of train val test games per season split

I trained XGBoost classifiers (Chen & Guestrin, 2016) to predict whether the team represented by each observation won the game. I first compared several feature configurations, including all available features, redundancy pruned features, differential only features, and separate feature groups representing shooting, rebounding, scoring, defense, pace, and playmaking. Feature influence was evaluated using SHAP values (Lundberg & Lee, 2017) and partial dependence plots.

Based on these analyses, I retained seven statistical differentials: defensive rebounds, assists, points off turnovers, three-pointers made, free throws made, opponent points in the paint, and blocks. Defensive rebounds had the largest mean absolute SHAP value, followed by assists and points off turnovers. Model hyperparameters were tuned through a search procedure, and training used early stopping based on validation set log loss.

3.2 Stage 2: Player-Level Season Projection

For Stage 2, I collected player level data using `nba_api` (Swar, n.d.). The dataset incorporated information from box scores, shot locations, defensive tracking, shot frequency, and league schedules. The endpoints were cleaned and merged within each season, with players matched across sources using normalized names. Counting statistics were then converted to per game averages to create consistent player level inputs.

Because rookies have no prior NBA statistical history, their projected statistics were estimated using XGBoost regressors trained on NBA 2K attribute ratings (NBA 2K API, n.d.). Predictor variables included shooting, passing, defense, athleticism, height, and position. A

separate regressor was trained for each target statistic. When evaluated on a held-out 20% sample of veteran players, the models recovered the primary per-game statistics with R^2 values ranging from approximately 0.75 to 0.84, including points, assists, and field goals made.

I then constructed projected 2026–27 rosters using NBA.com roster information and ESPN projected depth charts. For each team, a nine man rotation was selected subject to positional constraints. Player minutes were rescaled to sum to 240 minutes per game, corresponding to the total available minutes across five positions and four quarters. Expected minutes and games played were subsequently adjusted to account for known injuries and anticipated changes in player roles.

The projected player statistics were aggregated to create team level box score projections. For each scheduled game, the projected statistics of the two teams were converted into team level differentials corresponding to the inputs used by the Stage 1 model. Each game was then scored using the Stage 1 model from both teams' perspectives. The resulting win probabilities were adjusted on the logit scale to account for home-court advantage and a back-to-back penalty for the team playing on the second night of consecutive games.

The season was simulated in chronological schedule order. As the simulation progressed, each player's remaining expected games were decremented to account for projected availability throughout the season. Predicted game results were accumulated into team win totals. The resulting win totals were then shifted two games toward .500 to account for the model's observed tendency toward extreme projected records. Finally, the projected team win totals were compared with preseason Vegas over/under lines using mean absolute error (MAE).

Together, the two stages create a pipeline from historical team performance to season win totals. This framework allows the project to evaluate not only which box score statistics are associated with winning, but also whether those relationships can be applied to projected rosters to estimate future team performance.

4. Results

4.1 Category-Specific Models (Raw Team Statistics)

Category	Accuracy	AUC	Log Loss	Brier Score	Most Important Feature
Defense	75.1%	0.8313	0.5041	0.1677	OppEffFG%
Shooting	71.7%	0.7917	0.5539	0.1871	3PM
Playmaking	65.8%	0.7101	0.6242	0.2171	Ast
Rebounding	64.6%	0.6935	0.6338	0.2216	DReb
Scoring	55.6%	0.5872	0.6803	0.2436	%Points_InPaint
Pace	48.3%	0.4693	0.6944	0.2506	Pace

Defense produced the strongest results of any raw statistic category model, driven primarily by opponent effective field goal percentage (mean |SHAP| = 0.882), points off turnovers (0.657), and opponent turnover rate (0.295), with traditional counting stats like steals and blocks contributing comparatively little. Shooting was the second strongest category, driven overwhelmingly by three-point makes (mean |SHAP| = 1.232). Pace performed at or below the level of random classification.

4.2 Category-Specific Models (Differential Features)

Category	Accuracy	AUC	Log Loss	Brier Score	Most Important Feature
Shooting	97.37%	0.9952	0.1194	0.0286	diff_FTM
Scoring	92.61%	0.9867	0.1510	0.0468	diff_PointsInPaint
Playmaking	78.86%	0.8735	0.4459	0.1455	diff_Ast
Rebounding	75.22%	0.8261	0.5114	0.1701	diff_Reb%
Defense	71.78%	0.7963	0.5449	0.1841	diff_Blkc
Pace	55.11%	0.5726	0.6851	0.2459	diff_Possessions

Replacing raw statistics with differential (team - opponent) statistics produced substantial gains in nearly every category, most dramatically in Shooting (71.7% to 97.37% accuracy) and Scoring (55.6% to 92.61% accuracy). Pace remained the weakest category by a wide margin in both feature representations, and within Pace, only differential possessions carried a meaningful SHAP contribution; differential estimated pace, raw pace, and pace per 40 had negligible importance.

4.3 PDP-Based Final Feature Selection

Applying the 25 percentage point PDP threshold to the top SHAP ranked differential features within each category narrowed the candidate set to the seven features listed in Section 3.4, including `diff_Ast` and `diff_PointsOffTO` among others. Notably, differential rebounds were excluded from the final model despite reasonable SHAP importance, because the relationship was judged too intuitive/direct relative to the threshold criteria used for the other features.

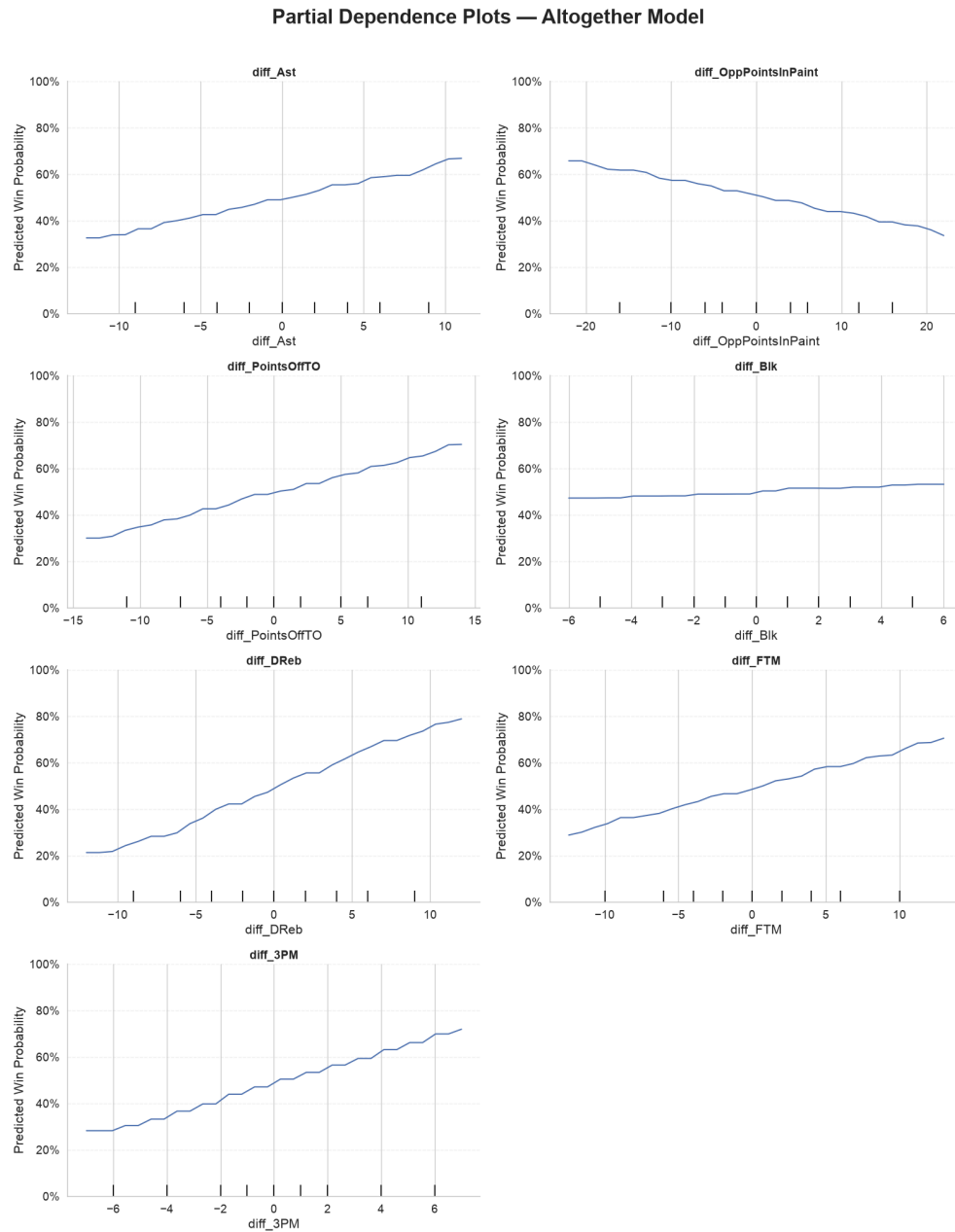


Figure 2: Partial Dependence Plots for each feature in the model

4.4 Final Combined Model

The final model, trained on the seven selected differential features, achieved:

Metric	Value
Accuracy	0.8868
AUC	0.9572
Log Loss	0.2645
Brier Score	0.0823

Confusion Matrix:

	Predicted Loss	Predicted Win
Actual Loss	3302	423
Actual Win	424	3334

The model correctly classified roughly 88.7% of games, with a balanced error rate between false positives and false negatives (423 vs. 424), indicating no systematic bias toward predicting wins or losses.

4.5 Season Simulation and Vegas Comparison

Western Conference

Team	Model W	L	Win %	Vegas O/U	Model - Vegas
SAS	63	19	.768	59.5	+3.5
OKC	59	23	.720	62.5	-3.5
POR	57	25	.695	42.5	+14.5
LAL	55	27	.671	46.5	+8.5
GSW	46	36	.561	40.5	+5.5
DEN	41	41	.500	49.5	-8.5
MIN	41	41	.500	48.5	-7.5
MEM	34	48	.415	28.5	+5.5
NOP	31	51	.378	27.5	+3.5
PHX	30	52	.366	38.5	-8.5
HOU	27	55	.329	47.5	-20.5
SAC	26	56	.317	21.5	+4.5
UTA	25	57	.305	35.5	-10.5
DAL	16	66	.195	34.5	-18.5
LAC	11	71	.134	30.5	-19.5

Figure 3: Projected Western Conference Standings and Vegas Odds for 2027 Season

Eastern Conference

Team	Model W	L	Win %	Vegas O/U	Model - Vegas
NYK	61	21	.744	52.5	+8.5
MIA	57	25	.695	46.5	+10.5
PHI	57	25	.695	50.5	+6.5
BOS	55	27	.671	51.5	+3.5
CLE	49	33	.598	47.5	+1.5
ORL	46	36	.561	43.5	+2.5
IND	45	37	.549	44.5	+0.5
ATL	42	40	.512	43.5	-1.5
WAS	42	40	.512	34.5	+7.5
CRA	41	41	.500	37.5	+3.5
DET	38	44	.463	49.5	-11.5
TOR	34	48	.415	45.5	-11.5
MIL	27	55	.329	25.5	+1.5
CHI	26	56	.317	27.5	-1.5
BKN	18	64	.220	24.5	-6.5

Figure 4: Projected Eastern Conference Standings and Vegas Odds for 2027 Season

Biggest positive deviations

Team	Difference
POR	+14.5
MIA	+10.5
NYK	+8.5
LAL	+8.5
WAS	+7.5
PHI	+6.5
GSW	+5.5
MEM	+5.5

Biggest negative deviations

Team	Difference
HOU	-20.5
LAC	-19.5
DAL	-18.5
DET	-11.5
TOR	-11.5
UTA	-10.5
DEN	-8.5
PHX	-8.5

Figure 5 and 6: Biggest positive and negative deviations when comparing with Vegas Odds

5. Discussion

The results support the project's central premise: relative, differential team performance is substantially more informative of victory than isolated, raw team statistics. This is consistent since a team's rebounding total means little without knowing how many rebounds the opponent secured. The near perfect accuracy of the differential Shooting and Scoring models should be interpreted with caution, however. Scoring and shooting differentials are close to a restatement of the game's outcome, so part of that accuracy is circular. The more informative result is that the compact final model, which excludes direct scoring measures, still classified outcomes with 88.7% accuracy.

The final model used seven differential features: assists (diff_Ast), opponent points in the paint (diff_OppPointsInPaint), points off turnovers (diff_PointsOffTO), blocks (diff_Blkc), defensive rebounds (diff_DRbc), free throws made (diff_FTM), and three-pointers made (diff_3PM). Together, these favor teams that share the ball, shoot well from three, protect the paint, finish defensive possessions with rebounds, and convert opponent mistakes into points. In roster terms, the model favors teams with elite top end talent and perimeter shooting: the three highest ranked teams in the projections were also those with the most star talent.

Pace, in both raw and differential form, showed little signal. Because both teams play essentially the same possessions in any given game, differential pace is close to zero by construction, so its weak performance is partly an artifact of how the feature is defined. Raw pace describes how fast a game was played rather than who won it, and teams win at both fast and slow tempos. This aligns with the broader basketball analytics literature, in which pace is typically treated as a context variable that affects counting stat totals rather than a driver of winning in its own right.

The selected features are partially consistent with Oliver's (2004) Four Factors framework, which emphasizes shooting efficiency, turnovers, offensive rebounding, and free-throw rate. Three-point makes capture an element of shooting efficiency, free throws made relates to the free-throw factor, and defensive rebounding relates to the rebounding factor. The mapping is loose, because the model uses counting statistics rather than the rate-based measures Oliver specifies, and offensive rebounding, one of the four factors, did not survive selection. The selected features also suggest that preventing high quality shots, through paint protection and blocks, and capitalizing on opponent mistakes, through points off turnovers, matters for winning, a finding with plausible coaching and roster construction implications.

The exclusion of offensive rebounding, despite meaningful SHAP importance in the category-specific model, illustrates the value of the two-stage SHAP + PDP selection process: importance to a model's predictions is not the same as a large, consistent relationship with win probability.

Finally, the projected win totals diverged from Vegas preseason lines by an average of 7.4 wins per team, and the model's projections were more extreme than the market's. The largest divergences were Houston (-20.5), the Clippers (-19.5), Dallas (-18.5), and Portland (+14.5). These gaps may reflect the rookie projections, ghost players, and carried-over minutes described below, but this was not tested directly. Because the comparison is against a market line rather than realized records, it shows where the model departs from consensus, not which is more accurate.

5.1 Limitations

Several limitations should be noted, and some of the modeling assumptions are admittedly strong. Most important, the classifier is trained on same-game box score differentials, so its accuracy reflects how well these statistics explain outcomes, not how well they predict them in advance. Season-level forecasts therefore depend on the accuracy of the projected per-game statistics, and any errors in those projections pass directly into the predicted win totals.

Those projections rest on several assumptions of their own. The rookie-projection model uses 2K ratings as a proxy for skill, which introduces uncertainty and potential bias inherited from how 2K assigns ratings, and because it was fit with logistic regression on current players, it may not capture how rookies actually develop or perform. Games played and minutes played are carried over from the prior season, so they ignore injuries, and role changes. On the off chance a roster is incomplete, they are filled with "ghost players," which keep the simulation running but introduce synthetic data that may not reflect actual replacement-level performance. The normalization procedure was likewise chosen on assumption rather than validated against alternatives, so results may be sensitive to that choice.

Finally, the simulation and modeling approach are both simplified. The season simulation accounts for only two schedule-based effects, back-to-backs and home-court advantage, and does not model trades, in-season roster changes, or coaching adjustments. The modeling approach was also not benchmarked against alternatives, and more sophisticated methods, such as gradient-boosted ensembles, hierarchical Bayesian models, or sequence-based player models, might yield better projections and are a natural direction for future work.

6. Conclusion

This project set out to answer what statistically constitutes "winning basketball," and to determine whether that understanding could be projected forward into a season level win prediction framework. The iterative XGBoost/SHAP feature selection process showed that differential, relative team performance substantially outperforms raw team statistics for predicting victory, with shooting efficiency (particularly free throws and three-point makes), paint scoring, assists, and defensive efficiency emerging as the most consistently influential drivers of winning, while pace showed minimal predictive value. The two-stage SHAP-then-PDP selection process yielded a compact, seven feature final model that classified game outcomes with 88.7% accuracy and strong probability calibration.

Beyond historical classification, the project's player-projection and roster-simulation pipeline demonstrates a path from individual player production to team-level, season-level outcome forecasting, including for players without existing NBA statistical history. Compared with Vegas preseason win totals, the projected records differed by an average of 7.4 wins per team (mean absolute difference across all 30 teams), with a slight net tendency to project below the market (-1.3 wins per team on average). The model landed within 3.5 wins of the Vegas line for 11 of 30 teams, most of them in the Eastern Conference (8 of 15, versus 3 of 15 in the West). The projections were also more spread out than the market: the model's top team (San Antonio, 63 wins) and bottom team (the Clippers, 11 wins) sit well outside Vegas's range for those teams. The largest disagreements were Houston (-20.5), the Clippers (-19.5), Dallas (-18.5), and Portland (+14.5). Because these are comparisons against a market line rather than realized records, they show where the model diverges from consensus, not which is more accurate.

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